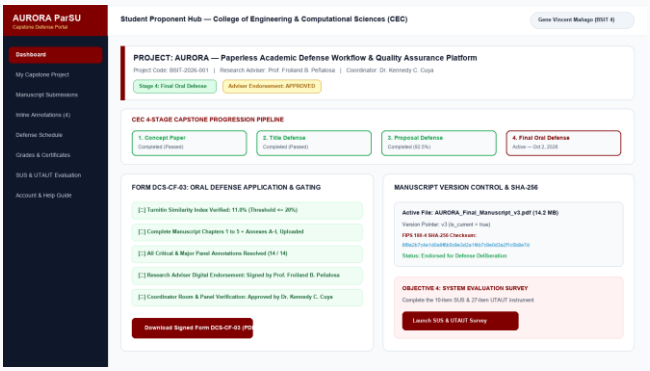
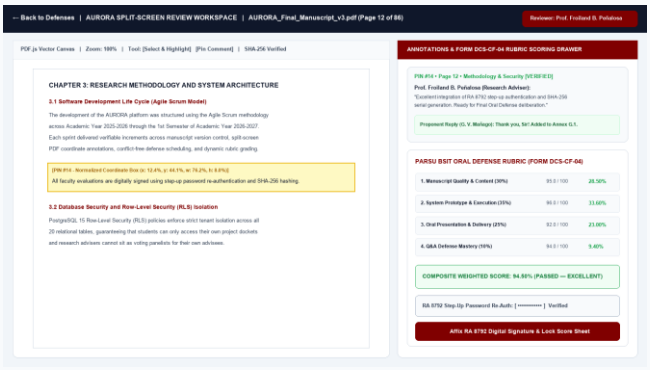
AURORA: PAPERLESS CAPSTONE DEFENSE MANAGEMENT PLATFORM

Academic Unified Review, Observation, Rating, and Assessment • Partido State University (CECS - DCS)

Transforming Manual Paper-Bound Thesis Routing into a Cryptographically Verified, Zero-Paper Cloud Workflow



Proponent Capstone Hub & 4-Stage Stepper



Dual-Pane Split-Screen PDF.js & Rubric Workspace

1. Dual-Pane PDF.js & Coordinate Annotations

Review manuscripts directly in the browser with resolution-independent bounding-box highlights (x%, y%, w%, h%), severity tags, and live WebSocket threads.

2. Conflict-Free Auto-Scheduler & Firewall

Eliminates double-booked defense rooms and panelist overlaps while enforcing academic impartiality by blocking advisers from grading their own advisees.

3. Dynamic Rubric & RA 8792 Digital Signatures

Automates Form DCS-CF-04 weighted computations (30/35/25/10%) with step-up password re-authentication and SHA-256 serials (AURORA-YYYY-NNNNNN).

4. Self-Healing Maintenance & 3-2-1 DR Vault

Built-in administrative console (/admin/system-health) with 6 live diagnostic probes, one-click self-healing repair, and SHA-256 signed JSON cold backups.

EMPIRICAL IMPACT & QUALITY BENCHMARKS

- SUS Usability Score: 91.8 / 100 (Grade A — Excellent)
- UTAUT Acceptability Mean: 4.86 / 5.00 (Highly Acceptable)
- Review Turnaround Reduction: 18.4 days → 2.6 days (−85.9%)
- Proponents: G. V. Mañago, C. A. Castillo, R. M. Buenafe, L. M. Encinas
- Research Adviser: Prof. Froiland B. Peñalosa | Chair: Dr. Kennedy C. Cuya

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